PRODUCT REQUIREMENT DOCUMENT

OmniHub — Multiplatform Messaging & Automation Hub

Field	Details
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Frontend	React.js
AI Engine	Google Gemini
Database	PostgreSQL (Supabase Free Tier)

Final technical and operational requirements

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1. EXECUTIVE OVERVIEW & PRODUCT VISION

1.1 Problem Statement

Modern businesses interact with customers across diverse communication channels including WhatsApp, Instagram, Telegram, Discord, LinkedIn, and Website Live Chat Widgets. Managing these channels separately leads to critical operational bottlenecks:

- **High Response Delays:** Delayed responses to routine customer inquiries lead to missed business opportunities.
- **Repetitive Manual Work:** Support representatives spend excessive time answering recurring questions regarding pricing, setup instructions, and catalogs.
- **Risk of AI Misinformation:** Fully autonomous AI bots can generate inaccurate pricing or unverified offers, negatively impacting brand reputation.
- **Fragmented Customer Data & Invisible Metrics:** Customer contacts are scattered without central CRM synchronization, and business owners lack visibility into channel traffic, message counts, and agent response performance across platforms.
- **Zero Cross-Channel Analytics:** Business owners cannot answer basic questions such as "How many messages arrived on WhatsApp today versus Instagram?" — leading to blind operational decisions.

1.2 Product Vision

OmniHub consolidates customer interactions from **six primary messaging channels** into a single central dashboard. The system integrates:

1. Automated post-service interactive button navigation.
2. Instant delivery of rich media assets such as PDF brochures, catalogs, and images.
3. Gemini AI auto-responses for open queries in **English, Hindi, and Hinglish**.
4. **Client Approval Mode (Human-in-the-Loop)** allowing business owners to review, edit, and approve AI replies before delivery.
5. An **Auto-Pilot Switch** enabling seamless transition between automated responses and direct human agent takeover.
6. **Real-time Graphical Analytics and Daily Channel Traffic Reports** providing visual insights into inbound messages versus outbound replies across all connected applications.
7. **Multi-Channel Daily Traffic Dashboard** — app-wise inbound and outbound counters with four interactive graphs (Bar, Donut, Line, Stacked).

2. GOALS & OBJECTIVES

2.1 Business Goals

#	Goal	Success Metric
G1	Demonstrate multi-channel message ingestion	Five or more channels receiving live messages during trial
G2	Prove AI auto-reply quality across three languages	Ninety percent or higher language match accuracy in demonstration
G3	Showcase Human-in-the-Loop approval	Client can approve, reject, or edit within five seconds
G4	Deliver real-time analytics value	Dashboard updates within two seconds of message arrival
G5	Prove free-tier feasibility	Zero infrastructure cost during trial

2.2 Technical Goals

#	Goal	Success Metric
T1	Sub-1.5 second auto-reply latency	Webhook to AI to Outbound under 1500 milliseconds
T2	Webhook signature validation	One hundred percent HMAC-SHA256 verification for Meta platforms
T3	Data normalization	All platforms to Universal Schema in under 200 milliseconds
T4	Real-time dashboard	Socket.io push, no page refresh required
T5	Zero-downtime analytics	Counters increment atomically

3. TARGET AUDIENCE / USERS

Persona	Role	Primary Need
Business Owner	Client (decision maker)	See all channels in one place, approve replies, view analytics
Support Agent	Human-in-the-loop	Takeover conversations, polish AI drafts, tag leads
Operations Manager	Analytics consumer	Daily traffic reports, peak hour insights, export CSV or PDF
Developer (Intern)	Builder	Clean codebase, clear module separation, free-tier deployable

4. TRIAL VS PRODUCTION SCOPE

Aspect	Trial Phase (Current)	Production Phase (Future)
Channels	Telegram, Discord, Website Chat (Live) plus WhatsApp, Instagram (Sandbox or Mock)	All six channels live with verified business accounts
AI	Gemini Free Tier (15 requests per minute)	Gemini Pro or Paid Tier with higher throughput
Database	Supabase Free Tier (500 MB)	AWS RDS PostgreSQL with Read Replicas
Hosting	Render Free plus Vercel Free plus Ngrok	AWS ECS or GCP Cloud Run
Analytics	Daily and Hourly counters (basic)	Predictive analytics with ML forecasting
Approval Workflow	Basic pending queue	Full audit trail with SLA tracking
LinkedIn	Skipped (Partner Program pending)	Full integration post-approval

5. AUTHENTICATION & USER ONBOARDING

This module handles dashboard login, OTP verification, and user account management.

5.1 Admin Registration & Login

- **Email and Password registration** — Admin creates their own account
- **Email verification** — A verification link is sent to the registered email after signup
- **Password policy** — Minimum eight characters, one uppercase letter, one number, one special character
- **Login with Email and Password** — A JWT token is generated
- **Remember Me option** — Session maintained for thirty days
- **Forgot Password** — A reset link is sent to the registered email
- **Password reset token** — Expires within fifteen minutes

5.2 OTP Verification (Two-Factor Authentication)

- **OTP after login** — A six-digit OTP is sent to the registered email
- **OTP expiry** — Five minutes
- **Resend OTP** — Thirty second cooldown, maximum three attempts
- **Maximum attempts** — Account locked for fifteen minutes after five incorrect OTP entries
- **Device binding** — Device is remembered after successful OTP verification
- **Backup codes** — Ten emergency codes are generated

5.3 Session Management

- **JWT Access Token** — Fifteen minute validity
- **Refresh Token** — Seven day validity, rotated on every refresh
- **Concurrent session limit** — Maximum three devices
- **Session timeout** — Automatic logout after thirty minutes of inactivity
- **Login audit log** — IP address, device, and timestamp saved for every login
- **Suspicious login alert** — Email alert on login from a new device or location

5.4 User Roles

- **Super Admin** — Full system access
- **Admin** — Dashboard access, can approve
- **Agent** — Handles conversations only
- **Viewer** — Read-only access, cannot make changes

6. UNIVERSAL INGESTION & DATA NORMALIZATION ENGINE

This engine receives incoming messages from every platform, validates their origin, and converts them into a common format.

6.1 Platform Webhook Receivers

Each platform has its own webhook endpoint:

Telegram Bot API Receiver

- Endpoint: `/api/webhooks/telegram`
- Receives text messages, button clicks, and voice clips
- Verified using Bot Token
- Free and the easiest setup

Discord Gateway Receiver

- Endpoint: `/api/webhooks/discord`
- Captures direct messages and channel messages
- Verified using Bot Token
- Free and unlimited

Instagram Graph API Receiver

- Endpoint: `/api/webhooks/instagram`
- Receives DMs, story mentions, and media replies
- Verified using Meta Webhook signature

WhatsApp Cloud API Receiver

- Endpoint: `/api/webhooks/whatsapp`
- Receives incoming messages, button clicks, and media receipts
- Verified using Meta Cloud API

LinkedIn Messaging Receiver

- Endpoint: `/api/webhooks/linkedin`
- Captures organization page messages
- Skipped during trial, added later

Website Live Chat Socket Engine

- Real-time bi-directional connection
- Connected to the website widget through Socket.io
- One hundred percent free, self-built

6.2 Webhook Security & Verification

- **Signature Validation** — X-Hub-Signature-256 verified using HMAC-SHA256 for Meta platforms (WhatsApp and Instagram)
- **Token Verification** — Authorization token checked for Telegram and Discord
- **Replay Attack Prevention** — Duplicate messages rejected using timestamp and unique ID
- **IP Whitelisting** — Webhooks accepted only from trusted IP addresses
- **Rate Limiting** — Maximum one thousand requests per minute per platform

6.3 Data Normalizer (Universal Schema)

Every platform sends messages in a different format. The normalizer converts them into a common format:

- **message_id** — System-wide unique identifier
- **platform** — Source channel (telegram, discord, and so on)
- **external_chat_id** — Platform's own chat identifier
- **sender_name** — User's name
- **sender_contact** — Phone or email, if available
- **message_type** — TEXT, BUTTON_CLICK, MEDIA, or VOICE
- **text_content** — Message text
- **media_url** — Link to media, if present
- **timestamp** — ISO-8601 UTC format

6.4 Message Deduplication

- **Unique message_id** — Every message's unique ID is saved in the database
- **Duplicate detection** — The same message arriving again is ignored
- **Idempotency** — Webhook retries do not cause double counting

7. INTERACTIVE MENU & RICH-MEDIA DISPATCH ENGINE

This module handles buttons, menus, and file delivery.

7.1 Post-Service Interactive Button Flows

Automatic Welcome or Thank You Message

- Triggered automatically after a service is completed
- A message such as "Thank you for choosing us!" is sent
- Three or four interactive buttons are attached

Primary Interactive Button Menu

- **Option 1: [How to Setup]** — Step-by-step setup guide with PDF link
- **Option 2: [View Services]** — Sub-menu showing packages and pricing
- **Option 3: [Contact Support or AI Query]** — User can type a custom query or request a human agent
- **Option 4: [Talk to Human]** — Direct agent handover

Nested Sub-Menu Routing

- Supports nested buttons up to three levels deep
- A [Back] button on every level
- An option to return to the [Main Menu]

7.2 Dynamic Media Attachment Dispatcher

- **Keyword Matcher** — If a user types "brochure", "catalog", "qr code", or "pricing", the file is sent automatically
- **Native File Dispatcher**
- PDF documents such as setup_guide.pdf and product_catalog.pdf
- High-resolution images such as payment_qr.png and office_location.jpg
- **Fallback Link Delivery** — If the file cannot be attached due to size limits, a secure HTTPS link is sent instead

7.3 Platform-Specific Button Rendering

- **Telegram** — InlineKeyboardMarkup (unlimited buttons)
- **Discord** — Message Components (five per row)
- **WhatsApp** — Interactive Buttons (maximum three)
- **Instagram** — Quick Replies (maximum thirteen)
- **Website** — Custom HTML buttons (unlimited)

8. MULTI-LINGUAL AI ENGINE & PROMPT ENGINEERING

This module generates AI responses to open-ended queries.

8.1 Multi-Lingual Capabilities

- **English** — Complete comprehension and fluent generation
- **Hindi (Devanagari)** — Responses in formal Hindi script
- **Hinglish (Roman Script)** — For example, "Account setup kaise karein?" receives "Account setup karne ke liye inn steps ko follow karein..."
- **Language Matcher Guarantee** — The AI always replies in the user's own language

8.2 Knowledge Base & Context Rules

- **Business Context Injection** — Business name, services, working hours, pricing, and asset links are supplied to the AI
- **Anti-Hallucination Guardrails**
- The AI cannot invent fake prices, offers, or promises
- If a query cannot be answered: "Main aapki query note kar raha hoon. Hamari team jald hi aapase contact karegi."
- **Response Length Limit** — Maximum three sentences
- **Tone Control** — Professional, friendly, and helpful

8.3 AI Response Generation Flow

1. The user's message is received
2. The language is detected
3. The message is sent to the AI with business context
4. The AI generates a reply
5. The reply is trimmed to three sentences
6. The client approval mode is checked
7. The message is sent once approved

8.4 AI Fallback Handling

- **Gemini API Down** — "Our AI service is busy. A support agent will assist you shortly."
- **Timeout** — Same fallback message
- **Rate Limit** — Queued for retry
- **Invalid Response** — Forwarded to a human agent

9. CLIENT DASHBOARD & APPROVAL WORKFLOW

This is a three-panel React dashboard where the client manages everything.

9.1 Unified Inbox UI

Left Panel (Chat List Directory)

- Active conversations sorted by latest activity
- Channel badges such as Telegram, WhatsApp, and Instagram
- Filter tabs: All, Pending Approval, Unread, Human Assigned
- Search bar for finding conversations
- Unread count badge

Center Panel (Active Conversation Window)

- Real-time chat feed
- Different colors for User, AI Bot, and Agent messages
- Message status indicators: Sent, Delivered, Pending Approval, Rejected
- Response draft area with a manual input box
- Action controls: Send, Approve, Edit, Reject

Right Panel (Lead Details & Controls Sidebar)

- Customer profile: Name, Platform ID, Phone, Email
- Auto-pilot switch [ON/OFF]
- Customer tags such as #HOT_LEAD, #PRICING_INQUIRY, #SETUP_ISSUE
- Conversation history
- Notes section

9.2 Client Approval Toggle (Human-in-the-Loop)

Toggle State = ON (Auto-Approval Mode)

- AI and automated responses are sent immediately
- No manual intervention required

Toggle State = OFF (Manual Approval Mode)

- Generated responses are held in the PENDING_APPROVAL queue
- Highlighted on the dashboard
- Action controls:
- **[Approve & Send]** — Send immediately
- **[Edit Draft]** — Edit the text before sending
- **[Reject]** — Cancel the message

9.3 AI Rephrase / Polish Tool

- A [Polish with AI] button in the manual editor
- An agent types a rough note such as "aaj office band hai kal baat karenge"
- The AI converts it into a professional message: "Thank you for reaching out. Our office is currently closed today. We will connect with you tomorrow."
- The original intent is preserved

9.4 Auto-Pilot vs Human Agent Takeover Switch

- An [Auto-Pilot: ON/OFF] switch in every conversation header
- **Takeover Mode (OFF)** — All automated triggers disabled for that thread
- **Auto Mode (ON)** — AI and menu responses enabled
- **Per-conversation control** — One thread handled by a human, another by AI

9.5 Message Status Indicators

- **Sent** — Message dispatched
- **Delivered** — Reached the platform
- **Read** — User has read it, if the platform supports it
- **Pending Approval** — Awaiting client approval
- **Rejected** — Rejected by the client
- **Failed** — Delivery error

10. REAL-TIME LEAD EXTRACTION & CRM SIDEBAR

This module automatically extracts customer details.

10.1 Real-Time Lead Parsing

- **Pattern Extraction Engine** — Every message is scanned for:
- Email addresses
- Ten-digit phone numbers
- Name introductions such as "My name is..." or "I am..."
- **Auto-population** — Extracted details appear immediately in the Right Sidebar
- **No page reload** — Updated in real time

10.2 Lead Display & Tagging

- **Customer profile** — Name, phone, email, platform ID
- **Manual tags** — Agents can add their own tags
- **Auto tags** — #HOT_LEAD, #PRICING_INQUIRY, #SETUP_ISSUE
- **Lead status** — New, Contacted, Qualified, Converted, Lost

10.3 CRM Integration (Future)

- **Export to CSV or Excel**
- **Webhook to external CRM**
- **API access for third-party tools**

11. WEBSITE CHAT WIDGET

This module embeds a chat widget on the client's website.

11.1 Widget Features

- **Floating chat button** — Bottom-right corner of the website
- **Customizable colors** — Client can set brand colors
- **Welcome message** — Automatic message when the widget opens
- **Pre-chat form** — Name, email, and phone (optional)
- **File upload** — Users can send PDFs or images
- **Typing indicator** — "Agent is typing..."
- **Offline mode** — Displays a form when no agent is available
- **Mobile responsive** — Works on mobile devices

11.2 Embed Code

The client pastes a single `<script>` tag into their website:

```
<script>
window.OmniHubConfig = {
apiKey: "CLIENT_API_KEY",
theme: "light",
position: "bottom-right",
welcomeMessage: "Hi! How can we help you today?"
};
</script>
<script src="https://omnihub.vercel.app/widget.js" async></script>
```

11.3 Widget Customization

- **Colors** — Primary, secondary, text
- **Position** — Bottom-right or bottom-left
- **Size** — Small, medium, large
- **Language** — English, Hindi, or Hinglish
- **Business hours** — "We're offline" message outside working hours
- **Avatar** — Agent's photo
- **Sound notification** — Plays sound on new message

11.4 Widget Analytics

- **Widget opens** — Number of times opened
- **Messages sent** — Number of messages received
- **Conversion rate** — Chats that became leads
- **Average session time** — Duration of chats

12. GRAPHICAL ANALYTICS, DASHBOARD CHARTS & PLATFORM METRICS

This module shows how many messages arrived on each app today and how many replies were sent.

12.1 Real-Time Daily Channel Volume Tracker

App-Wise Inbound & Outbound Counter

- For every platform:
- Inbound Count (messages received)
- Outbound AI Replies (sent by AI)
- Outbound Human Replies (sent by agents)
- Outbound Menu Replies (sent via menu)
- Pending Count (awaiting approval)

Date Range Selector

- Today
- Yesterday
- Last 7 Days
- Last 30 Days
- Custom date range

12.2 Interactive Graphical Visualization Suite

1. Multi-Platform Comparison Bar Chart

- Grouped side-by-side vertical bars
- Inbound versus Outbound for each platform
- Color branding: Green (WhatsApp), Magenta (Instagram), Blue (Telegram), Purple (Discord), Dark Blue (LinkedIn), Teal (Web Chat)

2. Channel Distribution Donut Chart

- Percentage share of total incoming traffic
- Example: 40% WhatsApp, 25% Instagram, 15% Telegram, 10% Web Chat, 6% Discord, 4% LinkedIn

3. Hourly Peak Traffic Line Graph

- Twenty-four-hour timeline
- Shows message spikes
- Helps identify peak hours
- Useful for staffing decisions

4. Response Breakdown Stacked Chart

- AI Auto-Replied
- Interactive Button Menu Dispatched
- Human Agent Approved or Sent
- Pending Approval

12.3 Top-Level KPI Scorecards

1. **Total Daily Messages Processed** — Inbound plus Outbound
2. **AI Resolution and Automation Rate** — Percentage of queries resolved without human intervention
3. **Pending Approval Bottleneck Counter** — Real-time badge
4. **Average Response Latency** — Target: under 1.5 seconds
5. **Active Conversations** — Live count
6. **Leads Generated Today** — Count

12.4 Export & Report Generation Engine

- **One-Click Export** — PDF and CSV formats
- **Daily Summary** — Full data for the day
- **Weekly Report** — Seven days of data
- **Monthly Report** — Thirty days of data
- **Scheduled Reports** — Automatic email (future)

12.5 Real-Time Dashboard Updates

- **Socket.io** — Updates without page refresh
- **Live counters** — Increment on every message
- **Auto-refresh** — Charts update every five seconds
- **Notifications** — Alert on new message

13. CONVERSATION ASSIGNMENT & AGENT MANAGEMENT

This module distributes conversations among agents.

13.1 Agent Roles

- **Admin** — Can view all conversations
- **Supervisor** — Team conversations
- **Agent** — Only assigned conversations
- **Viewer** — Read-only access

13.2 Assignment Methods

- **Manual Assignment** — Admin assigns to an agent
- **Auto Assignment** — Round-robin or load-based
- **Skill-based** — Language or product knowledge
- **Territory-based** — Based on region

13.3 Agent Workload Management

- **Maximum concurrent chats** — Five per agent (configurable)
- **Queue system** — Conversations waiting for an agent
- **Priority flag** — VIP customers get priority
- **Escalation** — Complex queries go to a supervisor

13.4 Agent Status

- **Online** — Available for chats
- **Busy** — Maximum chats reached
- **Away** — On break
- **Offline** — Not available

13.5 Agent Performance Metrics

- **Response time** — Average first reply time
- **Resolution time** — Time to close a conversation
- **Chats handled** — Total chats
- **Customer satisfaction** — Rating
- **Messages sent** — Total messages

14. CANNED RESPONSES & QUICK REPLIES

This module provides predefined replies.

14.1 Canned Responses

- **Predefined templates** — For common queries
- **Shortcut codes** — `/pricing`, `/hours`, `/location`
- **Variable support** — `{customer_name}`, `{business_name}`
- **Category-wise** — Pricing, Support, Sales, FAQ
- **Multi-language** — English, Hindi, Hinglish
- **Search** — Type to search
- **Usage tracking** — Which template was used how often

14.2 Quick Replies in Dashboard

- **Slash commands** — Type / to see the list
- **Keyboard shortcuts** — Ctrl+1, Ctrl+2
- **Recent replies** — Recently used
- **Favorites** — Starred responses

14.3 Canned Response Examples

- **Pricing:** "Our pricing starts from a base rate. Would you like a detailed quote?"
- **Hours:** "We are open Monday to Saturday, 10 AM to 7 PM."
- **Location:** "We are located at the given address. Here is the map link."
- **Setup:** "Here is your setup guide."
- **Support:** "Our team will connect with you within twenty-four hours."

15. MESSAGE SEARCH & FILTER

This module provides search and filter options in the dashboard.

15.1 Search Features

- **Global search** — Search across all conversations
- **Message content search** — Text and keywords
- **Customer search** — Name, phone, email
- **Date range** — Specific period
- **Platform filter** — Only WhatsApp, only Telegram
- **Status filter** — Pending, Sent, Failed
- **Tag filter** — #HOT_LEAD, #PRICING_INQUIRY

15.2 Advanced Filters

- **Unread only** — Only unread messages
- **Pending approval** — Awaiting approval
- **Assigned to me** — My conversations
- **Has media** — Messages with media
- **Has lead** — Lead extracted
- **Language** — English, Hindi, Hinglish

15.3 Sort Options

- **Latest first** — Default
- **Oldest first**
- **Most messages**
- **Longest pending**
- **Priority**

15.4 Saved Searches

- **Save filter** — For repeated use
- **Share filter** — With the team
- **Quick access** — In the dashboard sidebar

16. CONVERSATION NOTES & INTERNAL COMMENTS

This module provides internal notes.

16.1 Internal Notes

- **Agent notes** — Notes about the customer
- **@mentions** — Tag a team member
- **Timeline** — Chronological order of notes
- **Attachments** — Attach files
- **Visibility** — Internal only, customers cannot see them

16.2 Note Types

- **General note** — Any information
- **Follow-up** — Reminder
- **Issue** — Problem log
- **Resolution** — How it was fixed
- **Escalation** — Sent to supervisor

16.3 Collaboration

- **Team mentions** — @agent_name
- **Assignment** — Assign with the note
- **Notifications** — Alert on mention
- **History** — Who wrote what and when

17. BUSINESS HOURS & AWAY MESSAGES

This module configures working hours.

17.1 Business Hours Setup

- **Per day hours** — Different for each day of the week
- **Multiple shifts** — Morning and Evening
- **Holidays** — Specific dates
- **Time zone** — Auto-detected
- **Special hours** — Festivals and sale days

17.2 Away Message

- **Auto-reply** — Outside business hours
- **Custom message** — "We are currently offline. We will reply within twenty-four hours."
- **Expected response time** — "We will be back at 10 AM"
- **Emergency contact** — For urgent matters
- **Multi-language** — English, Hindi, Hinglish

17.3 Business Hours Logic

1. Message arrives
2. Business hours are checked
3. If open, AI or menu flow is triggered
4. If closed, away message is sent and the message is queued
5. Next business day, an automatic follow-up is sent

17.4 Holiday Mode

- **Manual toggle** — "We are on holiday"
- **Scheduled** — Pre-set dates
- **Custom message** — Holiday-specific
- **Auto-resume** — Normal operation after the holiday

18. MESSAGE TEMPLATES (WHATSAPP)

This module manages pre-approved templates for WhatsApp.

18.1 WhatsApp Template Requirement

WhatsApp applies a 24-hour rule:

- **24-hour window** — Free replies allowed for twenty-four hours after the user messages
- **Outside window** — A template must be used
- **Template approval** — Approval required from Meta

18.2 Template Categories

- **Marketing** — Offers and promotions
- **Utility** — Order updates and receipts
- **Authentication** — OTP and verification
- **Service** — Customer support

18.3 Template Management

- **Create template** — From the dashboard
- **Submit for approval** — Send to Meta
- **Status tracking** — Pending, Approved, Rejected
- **Version control** — Track updates
- **Multi-language** — A separate template per language

18.4 Template Variables

- `{{customer_name}}` — Customer's name
- `{{order_id}}` — Order ID
- `{{amount}}` — Amount
- `{{date}}` — Date
- `{{business_name}}` — Business name

18.5 Template Examples

- **Order Confirmation:** "Hi {{customer_name}}, your order {{order_id}} is confirmed. Total: {{amount}}."
- **Delivery Update:** "Your order {{order_id}} is out for delivery. Expected by {{date}}."
- **Payment Reminder:** "Hi {{customer_name}}, your payment of {{amount}} is due."

19. MESSAGE STATUS & DELIVERY RECEIPTS

This module tracks message status.

19.1 Status Types

- **Sent** — Message dispatched
- **Delivered** — Reached the platform
- **Read** — User has read it
- **Failed** — An error occurred
- **Pending Approval** — Awaiting client approval
- **Rejected** — Client rejected it

19.2 Platform-wise Status

- **Telegram** — Sent, Delivered, Read
- **Discord** — Sent, Delivered
- **WhatsApp** — Sent, Delivered, Read (blue tick)
- **Instagram** — Sent, Delivered, Read
- **Website** — Sent, Delivered, Read (via socket)

19.3 Status Webhook Handling

Each platform sends status webhooks:

- **WhatsApp** — `/api/webhooks/whatsapp/status`
- **Instagram** — `/api/webhooks/instagram/status`
- **Telegram** — Polled through the Bot API

On a status update:

1. The database is updated
2. A socket event is sent to the dashboard
3. The analytics counter is updated

19.4 Delivery Failure Handling

- **Retry logic** — Three attempts
- **Fallback** — Send email or SMS
- **Alert** — Red badge on the dashboard
- **Log** — Save the failure reason

20. RATE LIMITING & SPAM PREVENTION

This module prevents spam and abuse.

20.1 Rate Limiting

- **Per user** — Maximum ten messages per minute
- **Per IP** — Maximum one hundred requests per minute
- **Per API key** — Maximum one thousand requests per minute
- **Per platform** — Platform-specific limits

20.2 Spam Detection

- **Keyword filter** — Blocks abusive words
- **Link filter** — Blocks suspicious links
- **Pattern detection** — Repeated messages
- **Duplicate detection** — Ignores the same content
- **Bot detection** — Automated behavior

20.3 Abuse Handling

- **Warning** — First time
- **Temporary block** — Fifteen minutes
- **Permanent block** — Repeated abuse
- **Report** — Alerts the admin

20.4 Content Moderation

- **Profanity filter** — Bad words
- **NSFW filter** — Adult content
- **Violence filter** — Threatening content
- **Custom filter** — Business-specific

21. API INTEGRATION & SYSTEM ARCHITECTURE

21.1 System Architecture Overview

Channel Layer

- Messages arrive from six platforms
- Received through webhooks or Socket.io

Ingestion Layer

- Signature verification

- Rate limiting
- Replay attack prevention

Normalization Layer

- Conversion to Universal Schema
- Storage in the database

Processing Layer

- Analytics counter increment
- Menu matcher check
- AI engine call

Approval Layer

- Client approval mode check
- Auto-send or pending queue

Output Layer

- Sent back through the platform API
- Socket event sent to the dashboard

21.2 API Endpoints Summary

Webhook Endpoints

- `/api/webhooks/telegram` — POST
- `/api/webhooks/discord` — POST
- `/api/webhooks/whatsapp` — POST
- `/api/webhooks/instagram` — POST
- `/api/webhooks/linkedin` — POST

Dashboard Endpoints

- `/api/conversations` — GET (list)
- `/api/conversations/{id}/messages` — GET (thread)
- `/api/conversations/{id}/takeover` — POST (human takeover)
- `/api/conversations/{id}/autopilot` — POST (auto on or off)

Approval Endpoints

- `/api/approvals/pending` — GET
- `/api/approvals/{id}/approve` — POST
- `/api/approvals/{id}/reject` — POST
- `/api/approvals/{id}/edit` — POST

Analytics Endpoints

- `/api/analytics/daily` — GET
- `/api/analytics/hourly` — GET
- `/api/analytics/export` — GET
- `/api/analytics/kpi` — GET

AI Endpoints

- `/api/ai/polish` — POST (rephrase draft)
- `/api/ai/generate` — POST (manual AI call)

Lead Endpoints

- `/api/leads/{conversation_id}` — GET
- `/api/leads/tags` — POST

21.3 Socket.io Events

- `new_message` — A new message arrived
- `analytics_update` — Counter update
- `pending_approval` — New approval request
- `conversation_update` — Conversation changed
- `typing` — User typing

22. DATABASE ARCHITECTURE & SCHEMAS

The database runs on **PostgreSQL** (Supabase free tier).

22.1 Core Tables

conversations Table

- id, platform, external_chat_id, sender_name, sender_phone, sender_email, auto_pilot, created_at, updated_at
- Unique constraint: (platform, external_chat_id)

messages Table

- id, conversation_id, platform, sender_type (USER, AGENT, BOT), direction (INBOUND, OUTBOUND), text_content, media_url, status, created_at

daily_platform_analytics Table

- id, date, platform, inbound_count, outbound_ai_count, outbound_agent_count, outbound_menu_count, pending_count, created_at
- Unique constraint: (date, platform)

hourly_analytics Table

- id, date, hour, platform, inbound_count, outbound_count
- Unique constraint: (date, hour, platform)

pending_approvals Table

- id, conversation_id, message_id, proposed_text, status, created_at

leads Table

- id, conversation_id, full_name, phone_number, email_address, lead_tags, created_at

users Table (Dashboard)

- id, email, password_hash, role, status, last_login, created_at

audit_logs Table

- id, user_id, action, entity_type, entity_id, old_value, new_value, ip_address, created_at

22.2 Data Integrity

- Foreign key constraints
- Unique constraints
- Check constraints for positive amounts and valid dates
- Triggers for auto-updating timestamps
- Transaction isolation for financial operations

22.3 Scalability Design

- Read replicas for analytics
- Connection pooling (PgBouncer)
- Partitioning for large tables (messages, analytics)
- Auto-scaling based on CPU

23. SECURITY, COMPLIANCE & EXCEPTION HANDLING

23.1 Security Specifications

- **Data Encryption at Rest** — AES-256-GCM
- **Data Encryption in Transit** — TLS 1.3
- **Authentication** — JWT with strict expiration
- **Webhook Validation** — HMAC-SHA256
- **Rate Limiting** — One thousand requests per minute per API key
- **Input Validation** — SQL injection and XSS prevention
- **CSRF Protection** — Token-based
- **File Upload Validation** — Type, size, and malware scan

23.2 Compliance

- **Nepal IT Act 2061** compliance, if deployed in Nepal
- **GDPR-ready** — Data export and right to erasure
- **PCI-DSS** — Payment card data (future)
- **Data Retention Policy** — Auto-deletion after one year

23.3 Exception & Edge Case Handling Matrix

Meta Webhook Timeout (over 15 seconds)

- Risk: High
- Impact: Meta marks the endpoint unresponsive
- Mitigation: Immediate HTTP 200 OK and asynchronous processing

Gemini AI API Downtime

- Risk: Medium
- Impact: AI auto-reply fails
- Mitigation: Send a fallback message

Expired OAuth Access Token

- Risk: High
- Impact: Outbound replies fail with HTTP 401
- Mitigation: NEEDS_REAUTH flag and dashboard alert

Unsupported Media Format

- Risk: Low
- Impact: Render error

- Mitigation: Save the media URL and show a download link

Duplicate Webhook Delivery

- Risk: Medium
- Impact: Double counting
- Mitigation: Unique constraint on message_id

Database Connection Exhausted

- Risk: High
- Impact: All requests fail
- Mitigation: PgBouncer and retry logic

User Sends Abusive Content

- Risk: Medium
- Impact: Brand reputation
- Mitigation: Content filter and auto-flag

AI Generates Wrong Price

- Risk: High
- Impact: Business loss
- Mitigation: Anti-hallucination guardrails and approval mode

24. BACKUP & RESTORE

This module handles data backup.

24.1 Backup Strategy

- **Daily backup** — Automatic
- **Weekly full backup** — Complete database
- **Monthly archive** — Long-term storage
- **Point-in-time recovery** — Restore to any time

24.2 Backup Storage

- **Supabase backup** — Daily on the free tier
- **Manual export** — CSV or JSON
- **Cloud storage** — Google Drive, Dropbox
- **Retention** — Thirty days daily, one year monthly

24.3 Restore Process

- **Full restore** — Entire database
- **Partial restore** — Specific tables
- **Point-in-time** — Specific time
- **Testing** — Restore in the test environment

24.4 Data Export

- **User data** — GDPR compliance
- **Conversation export** — CSV or JSON
- **Analytics export** — PDF or Excel
- **Lead export** — CSV

25. MONITORING, ALERTING & LOGGING

This module monitors system health.

25.1 Uptime Monitoring

- **UptimeRobot** — Free, up to fifty monitors
- **Check interval** — Five minutes
- **Alert** — Email or SMS
- **Status page** — Public

25.2 Performance Monitoring

- **Response time** — API latency
- **Database queries** — Slow queries
- **Memory usage** — RAM
- **CPU usage** — Processor
- **Error rate** — Failed requests

25.3 Error Tracking

- **Sentry** — Free tier, five thousand events per month
- **Error grouping** — Similar errors grouped
- **Stack trace** — Debug information
- **User context** — Who was affected

25.4 Alerting

- **Email alerts** — Critical errors
- **Slack integration** — Team notifications
- **Dashboard alerts** — Visual badges
- **Escalation** — Auto-escalate

25.5 Health Check Endpoint

- `/api/health` — System status
- Database connectivity
- AI service status
- Platform API status
- Socket.io status

25.6 Logging

Log Types

- **Application logs** — Code execution
- **Access logs** — API requests
- **Error logs** — Exceptions
- **Audit logs** — User actions
- **Security logs** — Suspicious activity

Log Levels

- **DEBUG** — Detailed information
- **INFO** — General information
- **WARNING** — Potential issue
- **ERROR** — Error
- **CRITICAL** — Severe

Log Storage

- **Render logs** — Free, seven days
- **File logs** — Local storage
- **External** — Logtail or Papertrail (free tier)
- **Retention** — Thirty days

26. CI/CD PIPELINE

This module handles automatic deployment.

26.1 GitHub Actions

- **Free tier** — Two thousand minutes per month
- **Auto test** — On every push
- **Auto deploy** — On the main branch
- **Rollback** — On failure

26.2 Pipeline Stages

1. **Lint** — Code quality
2. **Test** — Unit and integration
3. **Build** — Docker image
4. **Deploy** — Render or Vercel
5. **Smoke test** — Basic checks

26.3 Environments

- **Development** — Local
- **Staging** — Test environment
- **Production** — Live

26.4 Deployment Strategy

- **Blue-green** — Zero downtime
- **Canary** — Gradual rollout
- **Rollback** — Instant revert

27. TESTING, QA & ACCEPTANCE CRITERIA

27.1 Testing Types

- **Unit tests** — Individual functions
- **Integration tests** — Module interactions
- **End-to-end tests** — Full user journey
- **Load tests** — Performance
- **Security tests** — Vulnerabilities

27.2 Testing Tools

- **pytest** — Python testing
- **Jest** — React testing
- **Playwright** — End-to-end
- **Locust** — Load testing
- **OWASP ZAP** — Security

27.3 Test Coverage

- **Target** — Eighty percent coverage
- **Critical paths** — One hundred percent
- **Report** — Coverage report

27.4 Test Automation

- **Pre-commit hooks** — Run tests
- **CI pipeline** — Auto test
- **Nightly tests** — Full suite

27.5 Acceptance Criteria

Test Scenario 1: Message Normalization and Metric Increment

- GIVEN an incoming payload
- WHEN the webhook receives it
- THEN it is parsed within 200 milliseconds, saved, and the counter is incremented

Test Scenario 2: Manual Approval Queue Hold

- GIVEN approval mode is OFF
- WHEN the AI generates a response
- THEN the status is PENDING_APPROVAL and the message is held without dispatch

Test Scenario 3: AI Polish Tool

- GIVEN a rough draft
- WHEN [Polish] is clicked
- THEN the professional text replaces the draft

Test Scenario 4: Real-Time Analytics Chart

- GIVEN active messaging
- WHEN the Analytics tab is opened
- THEN the Bar and Donut charts render accurate data

Test Scenario 5: Daily Analytics Counter

- GIVEN ten inbound and eight outbound messages
- WHEN the API is queried
- THEN the counts match

Test Scenario 6: Language Match

- GIVEN a Hinglish query
- WHEN the AI replies
- THEN the reply is in Hinglish

Test Scenario 7: Auto-Pilot Takeover

- GIVEN auto-pilot is OFF
- WHEN a user sends a message
- THEN no auto-reply is sent and the agent handles it

28. DOCUMENTATION

This module covers project documentation.

28.1 API Documentation

- **Swagger or OpenAPI** — Auto-generated
- **Postman collection** — Import ready
- **cURL examples** — Copy-paste ready
- **Authentication guide** — Step-by-step

28.2 User Guide

- **Getting started** — Setup guide
- **Dashboard tour** — Feature overview
- **FAQ** — Common questions
- **Video tutorials** — Screen recordings

28.3 Developer Guide

- **Architecture** — System design

- **Code structure** — Folder explanation
- **Contributing** — How to contribute
- **Deployment** — Deploy guide

28.4 Documentation Tools

- **MkDocs** — Static site
- **Docusaurus** — React-based
- **GitHub Wiki** — Simple
- **Notion** — Collaborative

29. DEVELOPER ONBOARDING & HANDOVER PLAYBOOK

29.1 Coding Standards

- **Modular code** — A separate file per platform
- **No modification** of existing handlers
- **Universal Schema** must be followed
- **Analytics counter** triggered on every webhook
- **AI prompts** only in ai_service
- **Socket events** centralized

29.2 Folder Structure

- **app/api/webhooks/** — Platform webhooks
- **app/api/dashboard/** — Dashboard APIs
- **app/services/** — Business logic
- **app/models/** — Database models
- **app/schemas/** — Pydantic schemas
- **app/utils/** — Helpers

29.3 Environment Variables

- Database URL
- Gemini API Key
- Telegram Bot Token
- Discord Bot Token
- Meta App Secret
- JWT Secret
- Socket.io CORS origins

29.4 Deployment Guide

- **Backend** — Render.com (free tier)
- **Frontend** — Vercel (free)
- **Database** — Supabase (free)
- **Local Tunnel** — Ngrok (free)
- **File Storage** — Supabase Storage (free)

30. CLIENT ONBOARDING FLOW

This module handles new client setup.

30.1 Onboarding Steps

1. **Sign up** — Email and password
2. **Email verify** — Verification link
3. **Business info** — Name, industry, hours
4. **Channel connect** — Telegram, Discord, and so on
5. **AI setup** — Knowledge base upload
6. **Menu setup** — Configure buttons
7. **Team invite** — Add agents
8. **Go live** — Start receiving

30.2 Onboarding Wizard

- **Step-by-step** — Guided setup
- **Progress bar** — Completion status
- **Skip option** — Can be done later
- **Help tooltips** — On every step
- **Video guide** — Screen recording

30.3 Knowledge Base Setup

- **Text input** — Business information
- **File upload** — PDF, DOCX

- **URL scraping** — From the website
- **FAQ builder** — Common questions and answers
- **Multi-language** — English, Hindi, Hinglish

30.4 Channel Connection

- **Telegram** — Paste the bot token
- **Discord** — Invite the bot
- **WhatsApp** — Connect the Meta app
- **Instagram** — Connect the Meta app
- **Website** — Embed code

30.5 Testing Before Go-Live

- **Test message** — Send one
- **AI reply check** — Response quality
- **Menu test** — Buttons work
- **Approval test** — Toggle check
- **Analytics test** — Counter update

31. CHANNEL-WISE API SETUP GUIDE

31.1 Telegram (Free, Easy)

- **Where to get it** — @BotFather on Telegram
- **What you get** — Bot Token
- **Steps** — /newbot, then name, then token
- **Webhook** — setWebhook URL
- **Cost** — One hundred percent free

31.2 Discord (Free)

- **Where to get it** — discord.com/developers
- **What you get** — Bot Token and App ID
- **Steps** — New App, then Bot, then Token
- **Invite** — OAuth2 URL
- **Cost** — One hundred percent free

31.3 Website Chat (Self-Built)

- **Where to get it** — Nothing required
- **Tech** — Socket.io and React widget
- **Steps** — Backend socket server plus frontend widget
- **Cost** — One hundred percent free

31.4 WhatsApp (Meta)

- **Where to get it** — developers.facebook.com
- **What you need** — Meta Developer Account
- **Free Tier** — One thousand conversations per month
- **Trial** — Test number with up to five verified numbers
- **Webhook** — /api/webhooks/whatsapp

31.5 Instagram (Meta)

- **Where to get it** — Same Meta Developer App
- **What you need** — Instagram Business Account
- **Free Tier** — Limited
- **Trial** — Sandbox or mock data

31.6 LinkedIn (Skipped in Trial)

- **Where to get it** — developer.linkedin.com
- **Problem** — Partner Program approval required
- **Trial** — Skip
- **Future** — Sales Navigator or Partner Program

31.7 Gemini AI (Free Tier)

- **Where to get it** — aistudio.google.com
- **What you get** — Free API Key
- **Free Tier** — Fifteen requests per minute, one million tokens per day
- **Cost** — Free for trial

32. BILLING & SUBSCRIPTION (FUTURE SCOPE)

This module is for future monetization.

32.1 Subscription Plans

- **Basic** — One channel, one thousand messages per month
- **Standard** — Three channels, ten thousand messages per month
- **Premium** — All channels, unlimited

32.2 Pricing

- **Monthly** — Base rate per month
- **Quarterly** — Ten percent discount
- **Yearly** — Twenty percent discount
- **Custom** — Enterprise

32.3 Payment Gateway

- **Razorpay** — India
- **Stripe** — International
- **Fonepay** — Nepal
- **UPI** — India

32.4 Billing Features

- **Auto-renewal** — Recurring
- **Invoice** — PDF download
- **Payment history** — All transactions
- **Upgrade or downgrade** — Prorated
- **Grace period** — Seven days
- **Suspension** — On non-payment

32.5 Usage Tracking

- **Message count** — Monthly
- **Channel count** — Active
- **Storage usage** — Files
- **API calls** — Rate limit
- **Overage charges** — Extra usage

33. TRIAL TIMELINE & MILESTONES

33.1 Eighteen-Day Phased Delivery

Phase 1: Setup (Day 1-2)

- FastAPI project initialization
- Supabase database setup
- Ngrok installation
- Project structure

Phase 2: Core and Telegram (Day 3-4)

- Webhook receiver
- Data normalizer
- Telegram bot integration

Phase 3: Discord and Website (Day 5-6)

- Discord bot
- Socket.io chat widget

Phase 4: AI and Menu (Day 7-9)

- Gemini integration
- Interactive buttons
- Rich media dispatch

Phase 5: Dashboard (Day 10-12)

- React three-panel UI
- Approval queue
- Real-time updates

Phase 6: Analytics (Day 13-14)

- Daily and hourly tables
- Recharts graphs
- KPI scorecards

Phase 7: WhatsApp and Instagram (Day 15-16)

- Meta test number
- Graph API integration

Phase 8: Polish and Demo (Day 17-18)

- Testing

- Bug fixes
- Demo preparation

33.2 Milestone Deliverables

- **Day 4** — Telegram live
- **Day 6** — Discord and Website live
- **Day 9** — AI and Menu working
- **Day 12** — Dashboard functional
- **Day 14** — Analytics complete
- **Day 16** — WhatsApp and Instagram sandbox
- **Day 18** — Final demo

34. SUCCESS METRICS & KPIs

34.1 Business Metrics

- Five or more channels live in trial
- Ninety percent or higher language match accuracy
- Client approval within five seconds
- Dashboard update within two seconds
- Zero infrastructure cost

34.2 Technical Metrics

- Sub-1.5 second auto-reply latency
- One hundred percent webhook signature validation
- Under 200 millisecond normalization time
- Real-time dashboard with no refresh
- Zero-downtime analytics

34.3 Trial Success Criteria

- Telegram, Discord, and Website live demo
- AI reply in English, Hindi, and Hinglish
- Interactive buttons working
- PDF and image delivery
- Approval workflow demo
- Analytics dashboard with four graphs
- Client sign-off for the production phase

35. RISKS & MITIGATION

35.1 Technical Risks

- **Meta API approval delay** — Use a test number during trial
- **Gemini rate limit** — Queue and retry
- **Ngrok URL change** — Fixed domain (paid) or re-setup
- **Database connection limit** — PgBouncer
- **Socket.io scaling** — Redis adapter (future)

35.2 Business Risks

- **Client expectation mismatch** — Clear trial versus production scope
- **LinkedIn unavailability** — Skip during trial and document
- **WhatsApp policy change** — Monitor Meta updates
- **AI hallucination** — Strict guardrails and approval mode

35.3 Mitigation Strategy

- Early client demo on Day 6
- Weekly progress updates
- Fallback for every external dependency
- Documentation for every module
- Testing at every phase

36. FUTURE SCOPE (PRODUCTION PHASE)

36.1 Phase 2 Features

- All six channels live with verified accounts
- LinkedIn Partner Program integration
- Predictive analytics with ML
- Advanced CRM with lead scoring
- Multi-tenant support for multiple businesses
- Mobile app (React Native)

- Voice message transcription
- Sentiment analysis
- Auto-translation into more languages

36.2 Scalability Roadmap

- AWS RDS PostgreSQL with read replicas
- Redis for caching and rate limiting
- Kubernetes for orchestration
- CDN for media delivery
- Elasticsearch for full-text search
- ML models for forecasting

36.3 Monetization (Future)

- Subscription plans (Basic, Standard, Premium)
- Per-message pricing
- API access for third parties
- White-label solution

37. APPENDIX

37.1 Glossary

- **PRD** — Product Requirement Document
- **Webhook** — HTTP callback for events
- **Socket.io** — Real-time bidirectional communication
- **JWT** — JSON Web Token
- **HMAC** — Hash-based Message Authentication Code
- **KPI** — Key Performance Indicator
- **RBAC** — Role-Based Access Control
- **CRUD** — Create, Read, Update, Delete
- **SLA** — Service Level Agreement

37.2 References

- Telegram Bot API Documentation
- Discord Developer Documentation
- Meta Graph API Documentation
- Gemini API Documentation
- FastAPI Documentation
- React Documentation
- Recharts Documentation

37.3 Change Log

- **v1.0.0** — Initial trial PRD
- **v1.1.0** — Analytics module added (future)
- **v2.0.0** — Production phase (future)

END OF PRD

OmniHub Trial Phase — Multiplatform Messaging & Automation Hub

This document contains the complete A-to-Z specification: from authentication to analytics, every module, every feature, every edge case, every API, and every security detail — all written in tier-2 English without any Hinglish words, and fully aligned with a zero-cost, free-tier stack.